Sushant Lokhande

+1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

Education

San Jose State University — San Jose, CA
Master of Science in Computer Science, GPA 3.8/4.0 — *Aug 2024 to May 2026*

- Coursework: Distributed Systems, Advanced Algorithms, Database Systems, Machine Learning

University of Mumbai — Mumbai, India
Bachelor of Technology in Computer Engineering, GPA 3.9/4.0 — *Jun 2019 to May 2023*

Experience

Graduate Research Assistant, ML and Distributed Systems — Aug 2025 to May 2026
San Jose State University — *San Jose, CA*

- Built a **Docker** WebTransport and Media-over-QUIC testbed and exercised **14** failure and recovery scenarios
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, and telemetry collection
- Reduced rebuffering by **19%** against a fixed-rate baseline through controlled adaptation experiments
- Raised malware-family classification to **80.2%** across 17 families with a soft-voting ensemble (+2.4 points)

Software Engineering Intern, Real-Time Learning Systems — Jun 2022 to Oct 2022
Turito

- Cut failures under disruption by **42%** (3.1% to 1.8%) by hardening **retries**, **timeouts**, and state transitions
- Reduced p95 CPU by **22%** by profiling hot paths and removing extra allocations, copies, and lock contention
- Reduced production diagnosis time by **35%** with structured metrics and targeted logs across about 12 regions
- Lowered network egress by **14%** by removing redundant processing and simplifying routing and payload handling

Software Engineering Intern — Nov 2021 to Jan 2022
Trivia Softwares

- Built a **Rust/Tokio** load-testing framework that drove about **25,000** concurrent sessions against backend services
- Improved throughput by about **31%** at equal load by exposing lock contention and serialized WebSocket writes
- Cut performance-investigation time by about **40%** with tooling built on Rust workers, Python, SQLite, and Flask

Projects

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, Redis, SQLite, SSE*

- Cut median latency from **759 ms to 44 ms** and modeled spend **76%** by serving 78% of requests from cache
- Added per-key **rate limiting**, provider **failover**, and usage and cost tracking behind an OpenAI-compatible API

Portcullis, Policy-Enforcement Proxy | *Rust*

- Centralized **authorization** and rate-limit checks in front of protected services with logged decisions

Proxima, Vector Search Engine | *C++17, AVX2, CMake, pybind11*

- Served **12,049 queries/sec** at 0.999 recall, **1.8x** faster than hnswlib, with 75% less traversal memory

Open Source

- **Corsair** (6 merged PRs): hardened **webhook signature verification** for 6 integrations so bad secrets **fail closed**
- **NVIDIA garak**, **Microsoft MarkItDown**, **Fission** (5 merged PRs): traceability, parsing, and debugging fixes

Technical Skills

Languages: Java, Python, C++17, Rust, TypeScript, SQL
Backend: FastAPI, Flask, REST APIs, gRPC, WebSockets, asyncio, Tokio, JWT authentication, rate limiting
Data: PostgreSQL, Redis, SQLite, MongoDB, Kafka, Temporal
Tools: AWS (EC2), Docker, Linux, Git, CI/CD, pytest, Locust, CMake